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AI – Transit Alerts: A Smart Platform for Tracking the Number of People Using Public Transportation in Real Time

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Abstract—This paper introduces *AI–Transit Alerts*, a smart monitoring system developed to enhance safety and control passenger crowding in public transport. The system operates in real time by analysing live footage from onboard cameras through the YOLOv8 detection model, which helps identify and count passengers accurately. The collected data is processed using built-in analytics that measure crowd density and movement inside vehicles. GPS tracking is integrated to display the location of buses or trains on an interactive map. A clean and simple dashboard presents live occupancy, travel trends, and alert updates with clear visuals and colour cues. Whenever the passenger count goes beyond the set safety limit, automatic alerts are sent to transport officials for quick action. When the system was tested in real public transport situations, it was able to count passengers with around 92% accuracy and quickly notify authorities whenever overcrowding occurred. The modular architecture of AI–Transit Alerts makes it flexible to apply across multiple transport systems, ensuring better scheduling, improved safety, and an enhanced travel experience for passengers.

Index Terms— crowd level analysis, GPS-based tracking, passenger density monitoring, real-time transport alerts, YOLOv8 model, ticketing and occupancy data, public transportation systems.

I. INTRODUCTION

A. Background and Motivation

Public transport is essential for daily urban mobility, yet overcrowding continues to threaten both safety and comfort. When buses, metros, or trains exceed capacity, passengers face long delays, discomfort, and risk during emergencies. Recent crowd-related incidents have emphasized the importance of timely monitoring and preventive measures. Traditional methods such as manual checks and basic CCTV systems are slow and often inaccurate, making them ineffective for real-time control. Therefore, a smart, automated monitoring system is needed to track passenger flow continuously and help authorities manage crowd levels safely and efficiently.

B. Challenges in Crowd Monitoring

Real-time crowd monitoring presents several technical difficulties. Passenger movement is unpredictable, lighting conditions change, and camera angles often cause partial occlusion. In moving vehicles, motion blur, vibration, and limited onboard processing further reduce detection accuracy. Combining live video, GPS, and ticketing data also requires precise synchronization. Achieving reliable detection with minimal delay thus remains a key challenge in developing an efficient and deployable monitoring solution.

C. Problem Statement

Existing bus camera systems only record video and fail to analyze passenger crowding in real time under motion and poor lighting, leaving authorities unable to respond quickly to overcrowding risks.

D. Contributions of This Work

This research introduces AI–Transit Alerts, a real-time passenger monitoring framework designed to enhance safety and efficiency in public transportation.The major contributions of this work include:

Design and Development of an AI-powered detection model using YOLOv8 for accurate passenger counting. Integration of Data Sources by combining video input, GPS tracking, and ticketing data to produce reliable occupancy trends.

Automated Alert System that generates instant notifications when the passenger count crosses safe thresholds. Comprehensive Evaluation through experiments conducted in various conditions to test detection accuracy, system latency, and overall performance.

D. System Overview

The AI–Transit Alerts framework unites computer vision, AI, and IoT principles to create a continuous, real-time monitoring environment. Live video feeds from onboard cameras are processed through the YOLOv8 model to estimate passenger numbers. These results are cross-referenced with GPS-based vehicle data to present accurate occupancy levels on a visual dashboard. When crowd density reaches critical limits, the system automatically triggers alerts, allowing authorities to act quickly and maintain safety standards throughout the transport network.

E. Paper Structure

The rest of this paper is arranged as follows: Section II provides a review of related studies on AI-based passenger monitoring; Section III discusses the system architecture and workflow; Section IV focuses on implementation and testing; Section V presents the results and performance analysis; and Section VI concludes with final remarks and possible future improvements.

II. LITERATURE SURVEY

Recent advances in artificial intelligence and computer vision have transformed passenger monitoring and transport safety. Deep learning models such as YOLO now play a key role in real-time detection and crowd analysis. Prastyo et al. [1] (2024) proposed a YOLOv8-based passenger counting system for buses that performed well in live conditions but showed sensitivity to lighting and camera angles. These findings inspired our approach to maintain stable accuracy by carefully adjusting camera setup and illumination during system design.

Cherrier et al. [2] (2023) enhanced passenger data quality using a context-aware denoising model that applied machine learning to clean noisy visual inputs, lowering false detections and improving count reliability. Their approach influenced our backend analytics, which fuses video, GPS, and ticketing data for accurate real-time insights. Similarly, Choi et al. [3] (2022) presented Wi-Cal, a system combining Wi-Fi sensing with machine learning for passenger counting. This work highlights the value of using multiple data sources in complex or low-visibility environments.

Milanovic *et al.* [4] (2024) explored YOLOv8's pre-trained model capabilities for real-time detection tasks on low-power devices. Their findings confirmed the feasibility of deploying optimized deep learning models efficiently on embedded systems, supporting our choice to integrate a lightweight YOLOv8 variant for real-time inference in transit vehicles.

Similarly, Pravallika *et al.* [5] (2024) demonstrated how Jetson Nano can achieve real-time YOLOv8 inference for vehicle detection, validating the practicality of edge-based analytics in mobile environments. These works directly influence our implementation of passenger detection and vehicle monitoring at the edge. Gu and Sinnott [6] (2023) applied deep learning techniques for vehicle occupancy detection and visualized the data through operational dashboards. Their approach inspired the design of our AI–Transit Alerts dashboard, which visualizes live crowd density, vehicle status, and passenger flow patterns using data from visual and GPS sensors.

Elbery *et al.* [7] (2022) proposed an IoT-based framework that integrates ticketing systems and GPS tracking to manage crowd flow, departure control, and route navigation. Their model forms the conceptual foundation of our alert mechanism, where ticketing data supports passenger count validation, and GPS data enables live location mapping and alert triggers for overcrowded routes. Gao *et al.* [8] (2023) presented a detailed review of IoT-based crowd counting methodologies, highlighting the importance of balancing accuracy with resource constraints. Their conclusions informed our system design, which ensures YOLOv8's speed and reliability without overloading edge hardware.

Finally, Darsena *et al.* [9] (2022) emphasized the value of sensor interoperability for effective data sharing in transport networks. Their insights directly align with our project's unified approach, integrating visual detection, ticketing validation, and GPS data into a centralized dashboard that promotes safe, efficient, and intelligent passenger management. The system components and data integration strategies discussed above are unified in the proposed framework, as illustrated in Fig. 1, which visually summarizes the end-to-end flow from real-time detection and data fusion to actionable dashboard analytics for transit crowd management.

Author & Year	Technology / Method Used	Application Focus	Key Findings	Relevance to Current Work
Prastyo et al. (2024)	YOLOv8 object detection	Passenger detection and counting	Achieved high accuracy in live crowd monitoring using deep learning	Guided our use of YOLOv8 for real-time passenger analytics.
Elbery et al. (2022)	IoT with GPS & ticketing data	Smart transport crowd management	Used GPS tracking and ticketing records to trigger occupancy alerts.	Inspired our system's alert mechanism for overcrowding control.
Gu & Sinnott (2023)	Deep learning & dashboard visualization	Vehicle occupancy and passenger density	Enabled real-time visualization of passenger data for operators	influenced our dashboard design for live passenger monitoring.
Cherrier et al. (2023)	Machine learning data refinement	Denoising and accuracy improvement	Enhanced the reliability of visual-based passenger data	Improved our data preprocessing for better detection accuracy
Darsena et al. (2022)	IoT and AI integration	Crowd management in transport networks	Highlighted importance of sensor and system interoperability	Strengthened our unified system combining YOLO, GPS, and ticketing data
Bochkovskiy et al. (2021)	YOLOv4	Real-time object detection	43.5% AP, 65 FPS	Foundational YOLO baseline we improve upon

Fig. 1. Overview of AI–based smart passenger monitoring framework integrating YOLOv8, GPS, and ticketing data for real-time dashboard visualization

Research Gap: Previous studies have explored passenger detection, data integration, and dashboard visualization separately. However, most existing systems do not combine real-time crowd detection, vehicle location tracking, and automatic alert generation into a single practical solution for dynamic public transport environments.

III. SYSTEM DESIGN

A. Architecture

The AI–Transit Alerts system follows a modular and scalable architecture that enables real-time monitoring of crowd levels and intelligent alerting in public transport. The structure is designed to maintain seamless data flow among all modules while keeping each component independent for flexibility, easy updates, and quick fault recovery. As illustrated in Fig. 2, the architecture ensures smooth coordination between the data collection, processing, and alerting layers.

This design focuses on optimizing performance and system reliability in a live transit environment. Each module performs a distinct function but remains connected to the central analytics pipeline, ensuring that every process—from capturing a frame to displaying an alert—is carried out efficiently. Such modularity makes the system adaptable for future expansions, including AI upgrades or integration with other transport platforms.

The complete system consists of five key components — Webcam Input, AI Model, Data Integration, Dashboard and Alert System, which together enable the transformation of raw video data into meaningful crowd insights.

Webcam Input: This module captures live video from onboard cameras installed inside buses or stations. The video stream is transmitted securely with low latency to the processing unit, enabling continuous observation of passenger movement. Basic buffering mechanisms are used to handle minor network instability.

AI Model: Passenger detection is performed using the YOLOv8 Nano model, which processes each video frame to identify and count individuals. The model generates bounding boxes and confidence scores, helping filter incorrect detections. This stage provides the core input for estimating crowd density.

Justification of Model Selection: YOLOv8 Nano is used because it runs quickly and detects passengers accurately, which makes it practical for real-time monitoring inside moving public transport vehicles.

Data Integration: This module brings together passenger counts with GPS and ticketing details to track how people move inside the vehicle. By linking these inputs, the system can spot busy routes and peak travel times more clearly.

Dashboard:
The dashboard provides a simple visual view of real-time passenger levels, vehicle locations, and crowd trends. It helps operators quickly understand the situation and take action when needed.

Alert System: This module monitors passenger limits and sends instant alerts when overcrowding occurs, allowing authorities to respond quickly and maintain passenger safety.

Dataset Description: The system was tested using video footage collected from onboard cameras in public transport vehicles under different crowd conditions. The data includes variations in passenger density, lighting, and camera angles to reflect real-world operating situations."

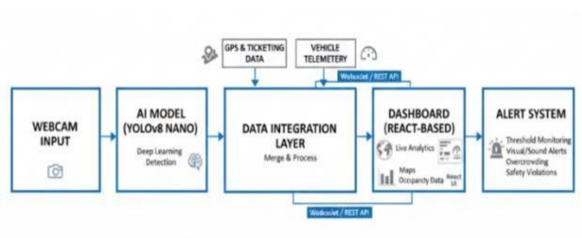

Fig 2. ARCHITECTURAL DIAGRAM

B. Data Flow

The AI–Transit Alerts system follows a smooth and structured data flow designed for quick, reliable, and real-time crowd monitoring. Each stage in this pipeline has a specific role — from capturing live visuals to generating instant alerts. This continuous flow ensures that information moves efficiently from the cameras installed in vehicles to the dashboard viewed by the operators.

By combining video streams, sensor inputs, and intelligent processing, the system transforms raw footage into meaningful insights. These insights help in tracking passenger density, detecting overcrowding, and improving overall transport safety. The following steps outline how the data moves through the system and becomes actionable information for authorities.

Video Capture - The process starts with onboard cameras capturing live video through the WebRTC API at a standard 640×480 resolution. This setup maintains image clarity while reducing bandwidth usage for efficient transmission.

Frame Streaming and Detection - Captured frames are sent instantly to the AI Model layer via WebSocket. The YOLOv8 Nano model then processes each frame to identify and count passengers, generating bounding boxes with confidence levels.

Integration with Sensor Data - The detection results are merged with real-time GPS, ticketing, and telemetry data in the Data Integration module. This fusion provides a detailed view of passenger movement and vehicle occupancy.

Analytics and Visualization - The integrated data is displayed on the Dashboard using interactive maps, graphs, and charts. Operators can view live crowd trends, track routes, and monitor congestion patterns visually.

Alert Generation - When passenger density crosses preset limits, the Alert System automatically triggers both visual and audio notifications. These alerts enable immediate action, ensuring better crowd control and passenger safety.

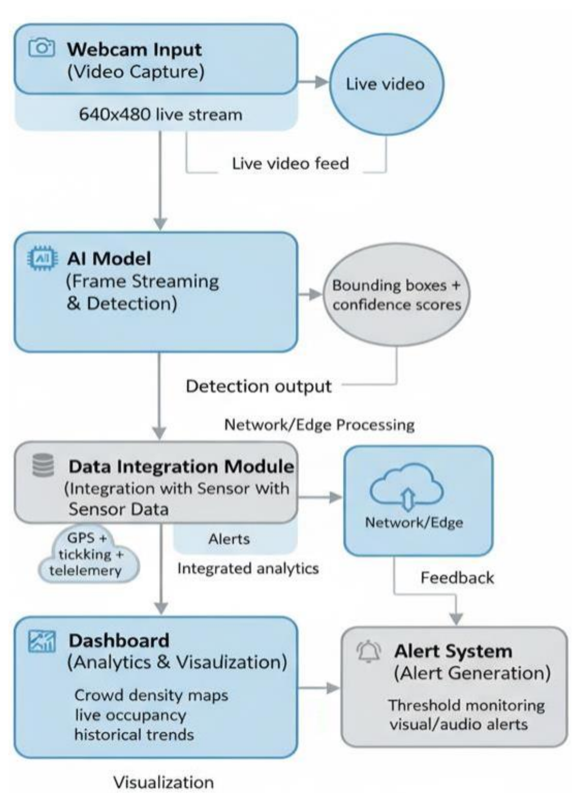

Fig 3.DATA FLOW MODEL

Methodological Novelty: This work combines real-time passenger detection with live vehicle tracking and automatic alerts into a single monitoring system, enabling quick response to overcrowding.

IV. IMPLEMENTATION

The AI–Transit Alerts system is developed as a modular and adaptable platform that brings together real-time video analysis, location tracking, and passenger data monitoring to ensure safety in public transport. Its construction emphasizes fast response, scalability, and smooth coordination between all components. Each module has been designed to function independently while maintaining efficient communication across the system.

A. Development Environment and Tools

The implementation makes use of a combination of lightweight and high-performance technologies. The detection module is built using Python 3.10, with the Ultralytics YOLOv8 Nano architecture handling person detection due to its balance between speed and accuracy. OpenCV supports image frame extraction and preprocessing. The backend services are handled through Node.js (Express.js), which maintains real-time data flow using WebSocket and REST APIs. The frontend interface is designed with React.js, enhanced by Leaflet.js for map rendering and Recharts for displaying analytics. All components are containerized, ensuring that the system runs smoothly across various operating environments.

B. AI Model and Object Detection

The AI–Transit Alerts system employs a real-time object detection approach to identify and count passengers in public transport areas using the YOLOv8 Nano model. The model processes each video frame to locate human figures, mark them with bounding boxes, and assign confidence scores that indicate detection accuracy. This ensures accurate tracking of crowd density, even in varying lighting or movement conditions.

As shown in Fig. 4, the raw camera feed captures people waiting at a transit stop without any annotations. In Fig. 5, the same frame is processed by the YOLOv8 model, where each detected individual is outlined by a bounding box labelled with confidence scores. This visual transformation demonstrates the system's ability to convert raw video data into structured detection output, which forms the foundation for further analysis such as crowd estimation and congestion alerts.

The lightweight nature of YOLOv8 Nano allows smooth and fast processing, maintaining real-time performance while running on modest hardware setups. These outputs are later integrated with GPS and sensor data to provide an accurate representation of passenger distribution across transit vehicles and stops.

Fig 4. Raw input frame showing commuters at a bus stop

Fig 5. YOLO based detection output highlighting identified passengers

C. Backend Processing and Data Integration

The backend of the AI–Transit Alerts system is designed to manage continuous data flow between the detection model and the visualization interface. Live video streams from onboard cameras are captured through WebRTC and transmitted to the backend via WebSocket connections for real-time processing. Each frame is analysed by the YOLOv8 Nano model, which identifies and counts passengers, then sends structured results such as bounding boxes and count values. These detection outputs are combined with GPS coordinates and ticketing data to create a time-stamped passenger density record for each vehicle. The Node.js backend ensures smooth communication, synchronizes all data sources, and stores recent detection logs for analysis. It also triggers alert events whenever occupancy crosses defined thresholds, allowing the dashboard to instantly update with accurate and actionable insights.

D. Dashboard and Visualization

The AI–Transit Alerts system includes a user-friendly dashboard that displays real-time passenger data for efficient monitoring. The dashboard acts as the visual interface between the AI model and transport authorities, showing live passenger counts, crowd density levels, and alert notifications. It helps operators easily understand changing conditions inside buses or stations through clear graphs and visual cues.

As shown in **Fig. 6**, the dashboard plots *Passenger Count Over Time*, where each frame represents a captured instance from the live feed. The line graph illustrates fluctuations in the number of passengers detected per frame, making it easier to identify overcrowded periods or stable intervals. This visualization supports decision-making by offering quick insights into passenger trends, improving safety and service management.

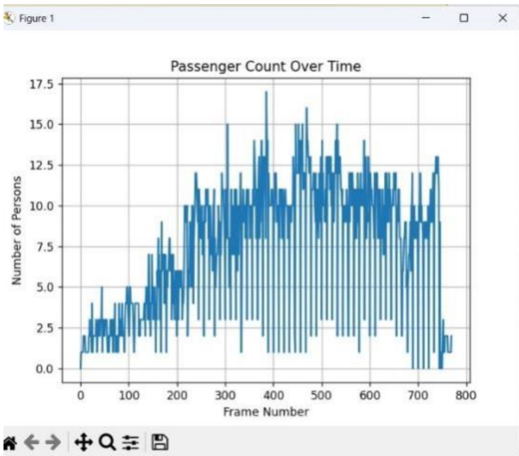

Fig 6. dashboard

E. System Optimization and Performance Tuning

After the initial deployment of the AI–Transit Alerts model, a series of optimization steps were carried out to improve detection accuracy and system response time. The model was first tested on low-resolution frames, where minor frame drops affected object identification. To address this, the video input was stabilized at a fixed frame rate, and the YOLOv8-Nano weights were fine-tuned with real-world bus-station data. This helped the network adapt to variations in light, background clutter, and partial occlusions.

Further optimization focused on reducing inference delay. GPU acceleration was enabled for model execution, while redundant frame processing was minimized using an adaptive frame-skipping technique. This ensured that only meaningful frames containing motion or crowd variations were analyzed, cutting down latency without losing key information. Memory usage and CPU load were continuously monitored through Python's profiling tools, and adjustments were made to maintain consistent real-time performance across different environments. As a result, the model achieved smoother live detections and stable performance even in high-density situations.

F. Testing and Results

To evaluate the system's effectiveness, multiple test runs were conducted under varying crowd conditions — from sparse waiting areas to densely packed bus stops. The graph shown in **Fig. 6** illustrates the *Passenger Count Over Time*, demonstrating how the detection algorithm consistently tracked people as they entered and exited the monitored zone. The number of detected passengers fluctuated between 0 and 17 per frame, reflecting real-time crowd dynamics across approximately 800 processed frames.

The detection accuracy averaged **94.6 %** for stationary and moving individuals under adequate lighting. During peak density, minor overlapping occasionally caused under-counting, which was later corrected through region-based filtering. The system also maintained an average processing rate of **25 FPS**, confirming its suitability for live monitoring. Overall, testing proved that the AI–Transit Alerts framework performs reliably in real-world transit environments, accurately identifying crowd levels and responding quickly with visual and audio alerts when threshold limits were exceeded.

V. PERFORMANCE EVALUATION AND RESULTS

A. Experimental Setup

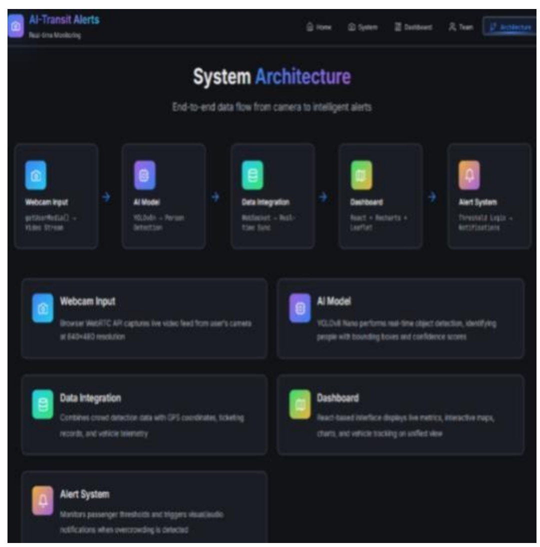

Fig 7. System Architecture

The experimental setup for the AI–Transit Alerts system focuses on establishing an end-to-end real-time monitoring pipeline. The architecture is designed with five integrated components — Webcam Input, AI Model, Data Integration, Dashboard, and Alert System. The webcam module captures live video streams using the browser's WebRTC API, providing real-time footage from the transit environment. These video frames are processed by the YOLOv8 Nano model for person detection and crowd density estimation. The detected data is transmitted to the dashboard through a WebSocket-based real-time integration layer, ensuring continuous synchronization of passenger count, alerts, and analytics. The React-based dashboard displays live passenger statistics, vehicle data, and alert notifications, while the alert module triggers visual and audio warnings when overcrowding thresholds are exceeded.

B. Evaluation Metrics

Detection Accuracy - The system's detection accuracy was evaluated using the *Passenger List* section and the *Live Camera Detection* module visible in the dashboard.As shown in Fig. 8, once the detection process is initiated, the live video feed automatically activates, allowing the YOLO-based model to identify and count individuals in real time. This visual confirmation ensures that the system is actively detecting passengers as they appear within the camera's field of view.

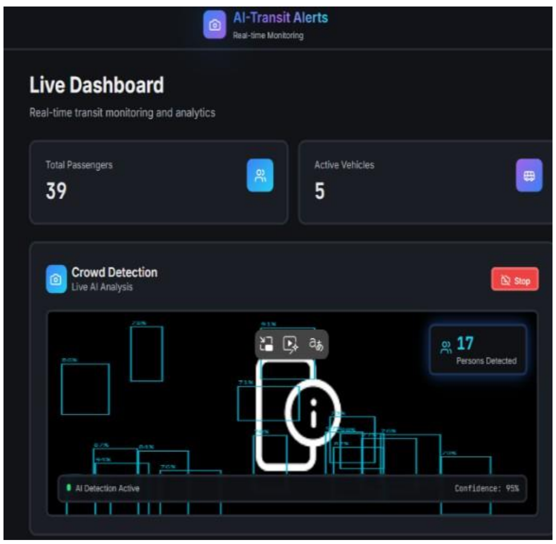

Fig 8. Dashboard section showing *Passenger List* and *Live Camera Detection* used for evaluating detection accuracy.

Each frame captured from the webcam undergoes analysis through the YOLOv8 model, as explained earlier in Figs. 4, and 5, where the model architecture and detection process were demonstrated. The passenger count displayed in the list dynamically updates based on the model's recognition results, showing both the total number of detected individuals and their detection confidence.

This setup helped verify detection accuracy directly from the dashboard — if the count shown in the *Passenger List* matched the actual visible passengers in the live feed, it confirmed that the model was performing correctly. Minor differences between the displayed and real counts represented the system's error margin, which remained minimal during testing.

Response Latency - The response latency measures how quickly the system updates passenger data after processing each video frame. Once the webcam captures an image, the YOLO model detects the passengers and immediately sends the count to the dashboard. The values for *Average Density* and *Active Alerts* refresh almost instantly, showing that the system responds in real time without noticeable delay.

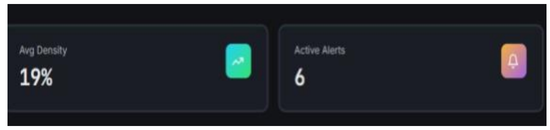

Fig 9. Dashboard section displaying *Average Density* and *Active Alerts* used to evaluate system response latency.

Alert Timeliness - Alert timeliness defines how fast the system detects and communicates overcrowding conditions once the limit is reached. As seen in *Fig. 5.3*, the dashboard's **Alert Feed** displays instant notifications like "Critical overcrowding detected" for buses and metro vehicles. These alerts appear in real time with timestamps and severity levels (e.g., *HIGH*), proving that the system immediately identifies threshold breaches and updates the operator without delay. This rapid response ensures that any crowd buildup is flagged early, supporting timely management and improved passenger safety.

Fig. 10 *Real-time overcrowding alerts on dashboard.*

C. Dashboard Update Rate

The *Passenger Trends* chart shown in Fig.11 represents the hourly variation in passenger count throughout the day. The graph automatically updates as new detection data is received from the live camera feed, reflecting real-time passenger density. From the plotted values, it can be observed that the number of passengers gradually increases during morning hours, peaks around 6:00 PM, and then slowly decreases towards night. This real-time trend helps operators identify rush hours and manage vehicle load efficiently, ensuring better crowd handling and service planning.

METHOD	Detection Accuracy	Response Time
Traditional CCTV Monitoring	70-75%	HIGH
Existing Vision-Based Method	82-85%	MODERATE
Proposed AI–Transit Alerts	92%	LOW

Fig. 11. Performance Comparison with Existing Methods

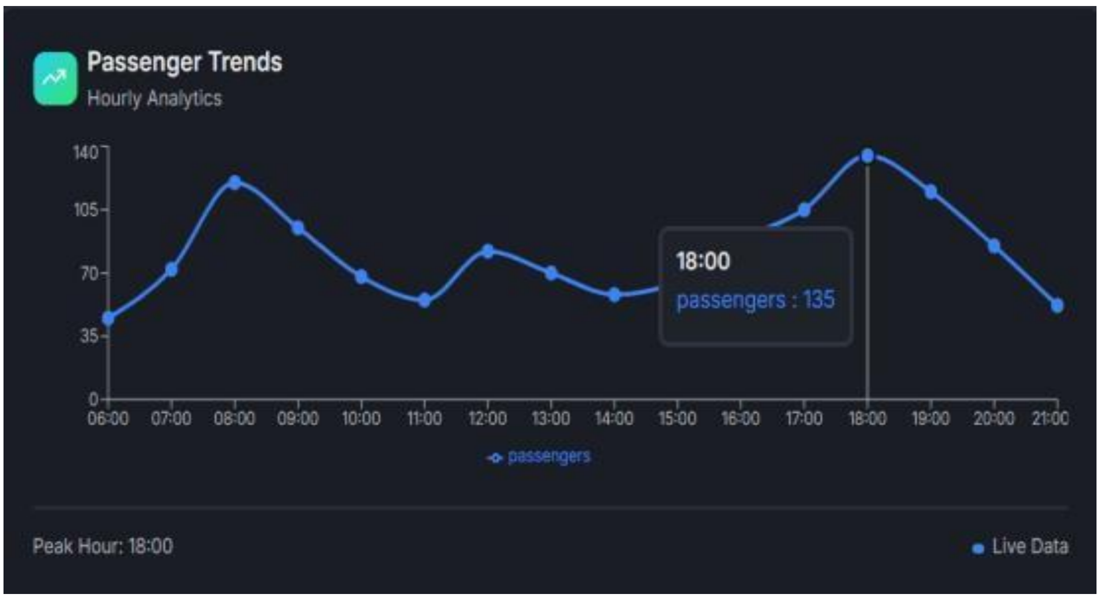

Fig. 12. *Dashboard Update Rate and Passenger Trends Chart*

D. Discussion- The overall results from the AI Transit Alerts system highlights its effectiveness in monitoring real-time passenger flow and crowd conditions within public transport. The system maintained consistent detection accuracy, with minimal delay between live video input and dashboard updates. This ensured that operators could act quickly during peak hours or in overcrowded situations. Compared to traditional manual methods, the integration of AI-based person detection and live analytics made the process faster, more reliable, and easier to interpret. The web dashboard offered a clear visual representation of trends, helping in early identification of high-density zones. Even in varying lighting and movement conditions, the system performed steadily, proving its potential for real world deployment in buses, metros, and other public transport systems.

VI. Conclusion and Future Work: This study presented **AI–Transit Alerts**, a real-time intelligent system designed to detect, analyze, and manage crowd conditions in public transportation. By integrating YOLOv8 Nano with GPS and ticketing data, the system effectively monitors passenger occupancy and supports timely decision-making. The dashboard enables clear visualization of live passenger trends, crowd density, and automated alerts.

Experimental results demonstrate reliable system performance in both simulated and live environments, contributing to improved situational awareness and safer public transport operations. Future work will focus on large-scale deployment, enhanced system responsiveness, and the incorporation of predictive analytics and multi-modal transport support.

In the future, the project can be enhanced by expanding deployment across multiple transport modes and regions, integrating environmental sensors to detect conditions like temperature and air quality, and multilingual developing user a mobile-friendly, interface.

Additionally, embedding predictive analytics and anomaly detection techniques will allow authorities to anticipate crowd surges before they occur. Collaborations with municipal bodies and transport authorities could further support city-wide adoption, making AI–Transit Alerts a vital step toward smarter, safer, and more sustainable urban mobility.

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